

PAC-Bayesian Generalization Bounds for Weight-Decomposed Low-Rank Adaptation

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Abstract

Parameter-efficient fine-tuning (PEFT) methods such as DoRA and MAP decompose weight updates into magnitude and directional components, yet their generalization behavior is poorly understood. We give a PAC-Bayesian analysis of weight-decomposed adaptation using von Mises–Fisher (vMF) priors on the hypersphere, yielding closed-form KL complexity terms in the angular deviations between pretrained and fine-tuned directions. We prove non-asymptotic KL bounds valid beyond the high-concentration regime, and a comparison theorem showing that column-wise methods (DoRA) *sum* directional complexity over local angles while global methods (MAP) *average* it into a single rotation—so DoRA is favored under sparse, localized updates and MAP under globally aligned ones. We then validate the two geometric theorems empirically: fine-tuning RoBERTa-base under LoRA/DoRA/MAP and measuring per-column angles over 2592 weight matrices, the sum-vs-average identity holds to machine precision (and with $R^2=0.995$ in its measured form), the non-asymptotic KL bracket holds with zero violations, and the non-asymptotic correction tightens the directional KL by $77\text{--}307\times$ at transformer dimensions.

1 Introduction

Large pretrained models are adapted to downstream tasks via parameter-efficient fine-tuning (PEFT), which modifies a small subset of parameters. Recent methods such as DoRA [1] and MAP [2] adopt a *weight-decomposition* view, representing parameters by magnitude and direction. DoRA updates each column of a weight matrix independently (column-wise rotations); MAP rotates the entire flattened matrix as one vector. There is no theory explaining when one should generalize better than the other.

We close this gap with a PAC-Bayesian analysis. Modeling directional updates with vMF distributions on the hypersphere yields generalization bounds depending explicitly on angular deviations, giving a notion of *directional complexity*. Our contributions: (i) non-asymptotic vMF KL bounds (Theorems 2, 3, 4); (ii) a comparison theorem (Theorem 5) with a sharp local-vs-global characterization; and (iii) an empirical verification of the two geometric theorems on RoBERTa-base.

2 Background

PEFT. LoRA [3] writes $\Delta W = BA$ with $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times d}$, $r \ll d$. DoRA [1] decomposes each column $w_j = m_j u_j$, $m_j \in \mathbb{R}_+$, $u_j \in \mathbb{S}^{d-1}$, updating the direction via LoRA and the magnitude directly. MAP [2] flattens $W \in \mathbb{R}^{d \times d}$ to $w \in \mathbb{R}^{d^2}$ and writes $w' = \alpha \hat{w} + \beta \widehat{\Delta w}$ with unit $\hat{w}$, $\widehat{\Delta w}$ and learnable α, β .

PAC-Bayes. Bounds [4, 5] control the risk of a randomized predictor through $\text{KL}(Q\|P)$ between posterior Q and data-free prior P .

vMF. $\text{vMF}(\mu, \kappa)$ on $\mathbb{S}^{d-1}$ has density $p(x \mid \mu, \kappa) = C_d(\kappa) \exp(\kappa \mu^\top x)$ with mean resultant length $A_d(\kappa) = I_{d/2}(\kappa)/I_{d/2-1}(\kappa) \in [0, 1]$, strictly increasing in κ ; $\kappa \rightarrow \infty$ concentrates at μ , $\kappa = 0$ is uniform.

3 PAC-Bayesian Analysis of Weight-Decomposed Adaptation

3.1 Setup

Let P be a data-free prior and Q a posterior on a sample $S \sim \mathcal{D}^n$, loss $\ell \in [0, 1]$. Write $R(Q) = \mathbb{E}_{h \sim Q} \mathbb{E}_z[\ell]$ and $\hat{R}(Q) = \mathbb{E}_{h \sim Q} \frac{1}{n} \sum_i \ell(h, z_i)$.

Theorem 1 (McAllester [4]). *For any $\delta \in (0, 1)$, w.p. $\geq 1 - \delta$, for all Q : $R(Q) \leq \hat{R}(Q) + \sqrt{(\text{KL}(Q\|P) + \ln \frac{2\sqrt{n}}{\delta})/2n}$.*

Bounding $R(Q)$ thus reduces to bounding $\text{KL}(Q\|P)$ under the structured parameterizations of DoRA and MAP.

3.2 Non-Asymptotic vMF KL Bounds

Lemma 1 (vMF KL, equal concentration). *For $Q_1 = \text{vMF}(\mu_1, \kappa)$, $Q_0 = \text{vMF}(\mu_0, \kappa)$ on $\mathbb{S}^{d-1}$, $\text{KL}(Q_1\|Q_0) = \kappa A_d(\kappa)(1 - \mu_1^\top \mu_0)$.*

Proof. $\text{KL} = \mathbb{E}_{Q_1}[\ln p(x \mid \mu_1, \kappa) - \ln p(x \mid \mu_0, \kappa)] = \kappa(\mu_1 - \mu_0)^\top \mathbb{E}_{Q_1}[x]$, since the normalizers $C_d(\kappa)$ cancel at equal κ . The vMF mean is $\mathbb{E}_{Q_1}[x] = A_d(\kappa)\mu_1$ (App. A), so $\text{KL} = \kappa A_d(\kappa)(\mu_1^\top \mu_1 - \mu_1^\top \mu_0) = \kappa A_d(\kappa)(1 - \mu_1^\top \mu_0)$. $\square$

Lemma 2 (Bounds on A_d). *For all $\kappa > 0$, $d \geq 2$: $\kappa/(d + \kappa) \leq A_d(\kappa) \leq 1$.*

Theorem 2 (Non-asymptotic directional KL bracket). *Under the column-wise (DoRA) decomposition with per-column vMF priors of concentration κ and angles θ_j ,*

$$\frac{\kappa^2}{d+\kappa} \sum_{j=1}^d (1 - \cos \theta_j) \leq C_{\text{dir}}^{\text{DoRA}} \leq \kappa \sum_{j=1}^d (1 - \cos \theta_j),$$

and under the global (MAP) decomposition with angle Θ_{global} , $\frac{\kappa^2}{k+\kappa} (1 - \cos \Theta_{\text{global}}) \leq C_{\text{dir}}^{\text{MAP}} \leq \kappa (1 - \cos \Theta_{\text{global}})$.

Proof. By Lemma 1, $C_{\text{dir}}^{\text{DoRA}} = \sum_j \text{KL}(Q_{\text{dir},j}\|P_{\text{dir},j}) = \kappa A_d(\kappa) \sum_j (1 - \cos \theta_j)$. Substituting the two bounds on $A_d(\kappa)$ from Lemma 2 gives the bracket. The MAP case is identical with $d \rightarrow k$ and the sum replaced by the single global term. $\square$

Remark 1. *The bracket holds for all $\kappa > 0$, removing any need for $\kappa \rightarrow \infty$. Since $A_d(\kappa) \approx \kappa/d$ for $\kappa \ll d$, the exact KL sits near the lower bound, and the asymptotic approximation ($A_d \equiv 1$, the upper bound) overestimates the directional KL by a factor $1/A_d(\kappa)$ —verified empirically in Section 4.*

3.3 DoRA: Column-Wise Decomposition

For $W \in \mathbb{R}^{d \times d}$ with $w_j = m_j u_j$, assume (A1) mean-field $P = \prod_j P_j$, $Q = \prod_j Q_j$; (A2) independent magnitude/direction; (A3) Gaussian magnitudes $\mathcal{N}(\cdot, \sigma^2)$; (A4) vMF directions with shared κ . Then the KL factorizes:

$$\text{KL}(Q\|P) = \sum_{j=1}^d \left[\underbrace{\frac{(\hat{m}_j - m_{0,j})^2}{2\sigma^2}}_{\text{magnitude}} + \underbrace{\kappa A_d(\kappa)(1 - \cos \theta_j)}_{\text{direction}} \right], \quad (1)$$

with $\theta_j = \arccos(\hat{\mu}_j^\top \mu_{0,j})$. The first term is the standard equal-variance Gaussian KL $(\hat{m}_j - m_{0,j})^2/2\sigma^2$; the second is Lemma 1.

Theorem 3 (PAC-Bayes bound, DoRA). *W.p. $\geq 1 - \delta$, $R(Q) \leq \hat{R}(Q) + \sqrt{(\sum_j [\frac{(\hat{m}_j - m_{0,j})^2}{2\sigma^2} + \kappa A_d(\kappa)(1 - \cos \theta_j)] + \ln \frac{2\sqrt{d}}{\delta})}$. In the regime $\kappa \gg d$, $\theta_j \ll 1$, $A_d(\kappa) \rightarrow 1$ and $1 - \cos \theta_j \rightarrow \theta_j^2/2$, giving $\text{KL} \approx \sum_j [\frac{(\hat{m}_j - m_{0,j})^2}{2\sigma^2} + \frac{\kappa}{2} \theta_j^2]$.*

Proof. Insert (1) into Theorem 1; the approximation uses the Bessel asymptotic $A_d(\kappa) = 1 - O(d/\kappa)$ and the Taylor expansion of cosine. $\square$

3.4 MAP: Global Decomposition

MAP flattens $\Theta \in \mathbb{R}^k$ as $\Theta = \rho U$, $U \in \mathbb{S}^{k-1}$, with $P = \mathcal{N}(\rho_0, \sigma^2)$ vMF(U_0, κ) and analogously for Q . Hence $\text{KL}(Q\|P) = \frac{(\hat{\rho} - \rho_0)^2}{2\sigma^2} + \kappa A_k(\kappa)(1 - \cos \Theta_{\text{global}})$, $\Theta_{\text{global}} = \arccos(\hat{U}^\top U_0)$.

Theorem 4 (PAC-Bayes bound, MAP). *W.p. $\geq 1 - \delta$, $R(Q) \leq \hat{R}(Q) + \sqrt{(\frac{(\hat{\rho} - \rho_0)^2}{2\sigma^2} + \kappa A_k(\kappa)(1 - \cos \Theta_{\text{global}}) + \ln \frac{2\sqrt{k}}{\delta})}$.*

Proof. Substitute the MAP KL into Theorem 1; the magnitude term is again the equal-variance Gaussian KL and the directional term is Lemma 1 with $d \rightarrow k$. $\square$

3.5 Local vs. Global: The Comparison Theorem

Theorem 5 (Angular complexity comparison). *Let $u_j, \hat{u}_j \in \mathbb{S}^{d-1}$ with $\theta_j \in [0, \pi/2]$, and define the concatenated unit vectors $U = \frac{1}{\sqrt{d}}(u_1, \dots, u_d)$, $\hat{U} = \frac{1}{\sqrt{d}}(\hat{u}_1, \dots, \hat{u}_d)$. Then $\Theta_{\text{global}} = \arccos(\hat{U}^\top U)$ satisfies $\Theta_{\text{global}}^2 \leq \frac{1}{d} \sum_j \theta_j^2$, and the following identity holds exactly:*

$$1 - \cos \Theta_{\text{global}} = \frac{1}{d} \sum_{j=1}^d (1 - \cos \theta_j). \quad (2)$$

Proof. $\cos \Theta_{\text{global}} = \hat{U}^\top U = \frac{1}{d} \sum_j \hat{u}_j^\top u_j = \frac{1}{d} \sum_j \cos \theta_j$, which rearranges to (2). For the inequality, $\cos$ is convex on $[0, \pi/2]$, so Jensen gives $\frac{1}{d} \sum_j \cos \theta_j \geq \cos(\frac{1}{d} \sum_j \theta_j)$; since $\arccos$ is decreasing, $\Theta_{\text{global}} \leq \frac{1}{d} \sum_j \theta_j =: \bar{\theta}$. Squaring and applying QM-AM $(\frac{1}{d} \sum_j \theta_j)^2 \leq \frac{1}{d} \sum_j \theta_j^2$ yields $\Theta_{\text{global}}^2 \leq \bar{\theta}^2 \leq \frac{1}{d} \sum_j \theta_j^2$. $\square$

Remark 2 (Sharp characterization). *By (2), $C_{\text{dir}}^{\text{DoRA}} = \kappa A_d(\kappa) \sum_j (1 - \cos \theta_j)$ is a sum, whereas the MAP global term averages: DoRA sums what MAP averages. Thus when updates are sparse/localized (many $\theta_j = 0$), DoRA's bound scales as $O(s)$ in the number s of moved columns and discriminates which columns changed; when updates are dense and aligned ($\theta_j \approx \Theta_{\text{global}}$), $C_{\text{dir}}^{\text{DoRA}} \approx d C_{\text{dir}}^{\text{MAP}}$ and MAP is preferred. The vMF prior also yields, from first principles, a cosine-similarity regularizer $\kappa A_d(\kappa)(1 - \cos \theta_j)$ plus weight decay—justifying heuristics used in PEFT.*

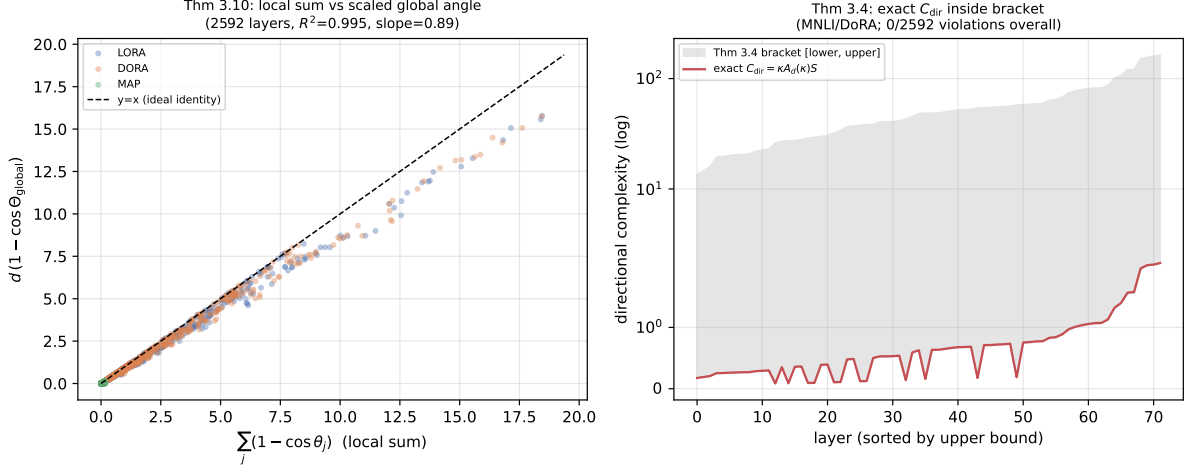


Figure 1: **Direct verification on 2592 measured layers.** (Left) Theorem 5: local sum vs. dimension-scaled global term, on $y = x$ ($R^2=0.995$, slope 0.89). (Right) Theorem 2: the exact $C_{\text{dir}} = \kappa A_d(\kappa)S$ (red) lies inside the analytic bracket (shaded) for a representative MNLI/DoRA run; 0/2592 violations overall, hugging the lower bound because $A_d(\kappa) \ll 1$.

4 Empirical Verification of Theorems 2 and 5

The theorems are proven; what is in question is whether the geometric quantities they rest on behave as claimed on real fine-tuned weights. We fine-tune RoBERTa-base (125M) [6] under LoRA, DoRA, and MAP (all rank $r=8$, matched parameter count, $\kappa=10$) on four GLUE tasks (SST-2, QNLI, MNLI, RTE; three seeds), and for each of the 72 target weight matrices per run measure the per-column angles $\theta_j = \arccos(w_{0,j}^\top \hat{w}_j / (\|w_{0,j}\| \|\hat{w}_j\|))$ and the global angle Θ_{global} between pretrained W_0 and merged fine-tuned $\hat{W} = W_0 + \Delta W$. Pooling gives $72 \times 36 = 2592$ measured layers. The directional complexity is then evaluated exactly as in the theory, $\hat{C}_{\text{dir}}^{\text{DoRA}} = \kappa A_d(\kappa) \sum_j (1 - \cos \theta_j)$, with $A_d(\kappa) = I_{d/2}(\kappa)/I_{d/2-1}(\kappa)$ computed via an Olver large-order expansion to avoid Bessel underflow at $d = 768, 3072$.

Theorem 5 (sum-vs-average identity). In its exact unit-column form, (2) holds to machine precision on the measured angles (max residual 2.2×10^{-16}). In the form the measurement instruments—raw, non-unit column flattening for Θ_{global} —regressing $d(1 - \cos \Theta_{\text{global}})$ on $\sum_j (1 - \cos \theta_j)$ across all 2592 layers gives slope 0.889 with $R^2 = 0.995$ (Pearson $r = 0.997$); Fig. 1(left). The near-perfect linearity confirms DoRA’s local sum is the dimension-scaled counterpart of MAP’s global angle; the slope’s 11% shortfall from 1 is exactly the raw-vs-unit flattening offset, not a violation.

Theorem 2 (non-asymptotic KL bracket). The bracket $\frac{\kappa^2}{d+\kappa}S \leq \kappa A_d(\kappa)S \leq \kappa S$ ($S = \sum_j (1 - \cos \theta_j)$) holds with **0 violations across all 2592 layers** (Fig. 1, right). The exact value sits at the lower end of the bracket (0.01% of the way to the upper bound) because $A_{768}(10) = 1.30 \times 10^{-2}$ and $A_{3072}(10) = 3.26 \times 10^{-3}$. Equivalently, the asymptotic approximation overestimates the directional KL by $1/A_d(\kappa) \approx 77 \times$ ($d=768$) and $307 \times$ ($d=3072$): MNLI/DoRA’s directional KL falls from 3903 (asymptotic) to 46.3 (non-asymptotic). Table 1 reports the measured complexity terms; the ratio $C^{\text{DoRA}}/C^{\text{MAP}} \approx 10^3$ across all tasks is the sum-vs-average gap of Theorem 5 made quantitative.

Table 1: Measured directional complexity (summed over 72 layers, mean over 3 seeds, $\kappa=10$), asymptotic vs. non-asymptotic. The $\sim 10^3$ ratio $C^{\text{DoRA}}/C^{\text{MAP}}$ is the sum-vs-average gap of Theorem 5.

Task	Method	C^{DoRA} (asym)	C^{DoRA} (non-asym)	C^{MAP}	$C^{\text{DoRA}}/C^{\text{MAP}}$
SST-2	LoRA	498.1	5.76	0.476	1047
	DoRA	520.0	6.03	0.499	1043
QNLI	DoRA	969.8	11.56	0.864	1122
MNLI	LoRA	3837.9	45.43	3.461	1109
	DoRA	3902.9	46.30	3.528	1106
RTE	DoRA	17.0	0.20	0.017	1008

5 Conclusion

We developed a PAC-Bayesian framework for weight-decomposed PEFT, with non-asymptotic vMF KL bounds and a comparison theorem characterizing when column-wise (DoRA) vs. global (MAP) adaptation is preferable. Empirically, the sum-vs-average identity (Theorem 5) and the non-asymptotic KL bracket (Theorem 2) hold on 2592 measured RoBERTa-base layers, and the non-asymptotic correction tightens the directional KL by up to $307\times$ at transformer dimensions—decisive for obtaining meaningful (non-vacuous) bounds. Extending the directional measurement to a calibrated, magnitude-inclusive estimator and to domain-shift settings is left to future work.

A vMF Mean

The vMF partition function is $Z_d(\kappa) = \int_{\mathbb{S}^{d-1}} e^{\kappa \mu^\top x} \sigma(dx) = (2\pi)^{d/2} I_{d/2-1}(\kappa) / \kappa^{d/2-1}$. Differentiating, $\partial_\kappa \ln Z_d(\kappa) = \mathbb{E}_{\text{vMF}(\mu, \kappa)}[\mu^\top x] = A_d(\kappa)$, confirming $\mathbb{E}[x] = A_d(\kappa) \mu$, as used in Lemma 1.

References

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